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Computational Neurosciences and AI

Structural Graph Characterization Of Narrative Disturbances In SSD During A Joint-task

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Declaration of originality

The present project contributes to a growing computational literature on language disturbances in schizophrenia spectrum disorders (SSD). While prior SpeechGraph studies have established that graph-theoretical metrics derived from spoken productions can discriminate individuals with SSD from non-clinical speakers and track formal thought disorder, this work has almost exclusively relied on monological tasks (free recall, dream reports, image description), conducted in isolation from any communicative partner. The structural signatures of narrative disturbances in SSD during a joint-task, and their sensitivity to graded referential complexity, remain largely uncharacterised.

The present study extends this work in three original directions: (1) it constitutes the first application of the SpeechGraphs framework to a joint-task (the StoryTelling in Sequence Task), moving beyond monological paradigms toward a setting that engages the pragmatic and referential demands of real communication; (2) it provides a direct test of the sensitivity of speech-graph structure to experimentally controlled referential complexity, using linear mixed-effects models to jointly estimate group differences and Group $\times$ Level interactions on the three primary metrics (LSCC, AD, ASPL); (3) it uses commonality analysis to quantify the unique variance in perceived communicative ease explained by graph structure beyond aggregate symptom severity.

Abstract

Background: People with schizophrenia spectrum disorders (SSD) present with narrative and pragmatic language impairments, yet most computational accounts have studied them in monological tasks. This study aimed to determine whether the structure of SpeechGraphs derived from spoken productions distinguishes SSD from community controls (CO) during a joint narration task, varies with referential complexity, and correlates with listener-rated perceived Facility ratings drawn from a prior study.

Methods: 50 participants with SSD and 68 CO performed the StoryTelling in Sequence Task (STST) with an interaction partner across three levels of referential complexity. From each transcript three SpeechGraphs metrics were computed. Linear mixed-effects models tested Group differences and interactions between Group and Referential Complexity. Pearson correlations related each metric to perceived Facility ratings provided by naïve listeners in a prior study; commonality analysis quantified the share of variance in Facility uniquely explained by graph structure compared to PANSS symptom.

Results: SSD productions yielded smaller strongly connected subgraphs than those of CO, and interaction between Group and Referential Complexity was observed for AD and ASPL. As Referential Complexity rose, SSD SpeechGraphs grew denser and structurally tighter compared to CO. Facility ratings are associated with different SpeechGraphs metrics for the two groups. Commonality analysis revealed that SpeechGraphs metrics explained a unique share of variance in Facility comparable to that of symptom severity.

Conclusions: SpeechGraphs structure carries a joint-task signature of SSD narrative impairments that scales with referential complexity demand, and captures dimensions of listener-rated quality not reflected in aggregate symptom ratings. These findings support structural descriptors of discourse production as biosocial markers of psychosis and motivate further extensions of SpeechGraphs use in clinical populations.

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1 Introduction

1.1 Literature-Review and Rationale

Individuals with schizophrenia spectrum disorders (SSD) exhibit a spectrum of cognitive and behavioral impairments (de Sousa et al., 2019; Heinrichs and Zakzanis, 1998), such as challenges with discourse production (Covington et al., 2005; de Boer et al., 2020; Docherty et al., 1996). Formal Thought Disorder (FTD) (Andreasen, 1979) is a common feature of various SSD (Passby and Broome, 2017), and FTD has been associated with altered patterns of social functioning (Marggraf et al., 2020). For instance, disconnected speech, a feature of positive FTD, has been shown to predict socially inappropriate behavior, above the effects of cognitive deficits and symptom severity (Bowie et al., 2011). Beyond their impact on social functioning, language disturbances can appear in individuals at clinical high risk for psychosis and may serve as early indicators of the disorder (Bearden et al., 2011; Elvevåg et al., 2007; Rezaii et al., 2019; Mota et al., 2017). Altogether, these findings suggest that a better characterization of the discourse production impairments exhibited by people with SSD is of interest for understanding the onset of the disorder, as well as for identifying potential barriers to recovery.

Language can be framed primarily as a communicative process (Fedorenko et al., 2024), in which the objective of language production would be the reduction of uncertainty for the listener (Ramscar and Baayen, 2013). Consistent with this view, there is increasing evidence suggesting that speakers adjust their discourse choices to the context and to the interaction partner (Clark and Wilkes-Gibbs, 1986; Gundel et al., 1993; Pickering and Garrod, 2021). For instance, Clark and Wilkes-Gibbs (1986) experimentally demonstrated that pairs of participants collaboratively refined their use of referential expressions (i.e. the terms they use to refer to a given entity) across repeated trials of a same task, shortening noun phrases as shared understanding accumulated. Building on such evidence, Pickering and Garrod (2004) proposed that alignment between interlocutors is the core mechanism underlying successful dialogue. Their Interactive Alignment Model (Pickering and Garrod, 2004) describes how interlocutors automatically align their situation models and linguistic representations through priming at lexical and syntactic levels, without requiring explicit modeling of the partner's mental states. Their later Shared Workspace Framework (Pickering and Garrod, 2021) broadens this account by embedding dialogue within a general theory of cooperative joint activity, adding the notion of meta-representation of alignment, whereby each participant monitors and adjusts to their perceived degree of mutual understanding with their partner. Together, these perspectives emphasize that language can be better understood through collaborative, communicative processes.

SSD are associated with discourse production impairment (Covington et al., 2005; de Boer et al., 2020; Docherty et al., 1996). FTD can manifest in discourse production as (1) derailment (abrupt shifts between unrelated ideas), (2) tangentiality (diverging from the main topic without addressing the question), (3) circumstantiality (providing excessive details before reaching the main point) and others (Andreasen, 1979). Beyond FTD, a recent meta-analysis of expressive pragmatic language in SSD identified moderate-to-large deficits across multiple pragmatic dimensions, including cooperativity, anaphora, cohesion, and coherence (Meister et al., 2026). As for referential expressions, individuals with SSD tend to use more first-person

singular pronouns and fewer first-person plural pronouns (Tang et al., 2021). More generally, these patterns suggest that discourse production deficits in SSD affect the overall organization, referential choices, and communicative uptake of discourse production.

Despite the fundamentally communicative nature of discourse production and the discourse production impairments observed in SSD, very few studies have used a task that requires the participant to verbally interact with an interaction partner to achieve a mutually shared communicative goal, namely, a joint-task. The few that have, consistently show that individuals with SSD show a reduced ability to adjust their discourse production to their interaction partner. Whether in their use of referential markers across trials (Champagne-Lavau et al., 2009), in their linguistic alignment in dialogue (Dwyer et al., 2020) or in the amount of information provided based on interaction partner's likely knowledge of the discussed material (Achim et al., 2022).

To allow for a more precise study of referential expressions use, two tasks were created to study how speakers use linguistic expressions that point to the different story characters across different stages of narrative elaboration during a joint-task (Achim et al., 2023; Fossard et al., 2018). Such as when introducing new story characters (typically with indefinite expressions), when maintaining characters in focus (typically with pronouns), or when reintroducing a characters that was previously introduced but was no longer in focus (typically with definite expressions) (Fossard et al., 2018; Sandoz et al., 2024). Both tasks used a similar paradigm in which participants narrated a story while their interaction partner attempted to reorder their own copy of the story images based on the narration, either from movies scenes (Achim et al., 2023), or six comic-strips in the StoryTelling in Sequence Task (STST; Fossard et al. (2018)). Primary results of Achim et al. (2025) suggest that individuals with SSD use fewer names and more definite markers at the introduction stage, with a reduced adjustment in definite marker use based on whether the interaction partner was likely to already know the characters being introduced. Regarding communicative uptake, individuals with SSD received lower Facility ratings from raters blind to group membership across both tasks (Achim et al., 2023; Cayouette et al., 2025), indicating that it was harder to reorder the images based on their narration. Notably, across all these studies, the measures used rely exclusively on manual coding or subjective ratings by naive listeners, such as word count, number of speaking turns, ratios of reference markers, or manually coded number and type of information pieces, and none incorporate a computational approach to analyze the verbal productions.

One way to go beyond manual approaches is Natural Language Processing (NLP) (Bird et al., 2009). NLP, the field of study at the crossroads of computer science and linguistics that uses automated analysis to understand natural human language. Reviews have highlighted the potential of NLP applied to schizophrenia. Corcoran and Cecchi (2020) describe how approaches such as semantic coherence analysis, syntactic complexity, and SpeechGraphs analysis can operationalize language disturbances in psychotic disorders. While a more recent scoping review by (Deneault et al., 2024) identified six major NLP applications in schizophrenia, ranging from diagnostic and predictive modeling to speech and communication analysis.

One such structural approach, particularly relevant to the quantification of discourse organization, is the SpeechGraph (Mota et al., 2012). A method to model spoken language as directed graphs (Bollobás, 1998)

in which words serve as nodes and their temporal succession defines edge directions. From these representations, structural metrics can be computed to quantify the structural organization of the discourse. SpeechGraphs analyses have been shown to discriminate participants with schizophrenia from Brazilian Portuguese participants with bipolar mania and non-clinical participants (Mota et al., 2012). Subsequent work with native English speakers replicated these findings, and further showed that connectivity of the discourse was associated with brain dysconnectivity, cognitive performance, and functional outcomes (Palaniyappan et al., 2019). Additionally, SpeechGraphs metrics have been shown to predict transition to psychosis in clinical high-risk populations, with various tasks, like memory reports (Mota et al., 2017) and image description (Spencer et al., 2021).

Yet a major observation cuts across all of these studies applying NLP to SSD research: speech samples were collected in monological settings such as dream reports, memory recall, picture description, or clinical interviews, where participants produced language without any shared goal to accomplish with a partner (Deneault et al., 2024) which can overlook the ecological validity of a joint-task and does not allow for examination upon communicative uptake. Furthermore, none of these paradigms affords experimental control over specific linguistic features such as referential complexity, thereby preventing systematic examination of how particular aspects of language production may modulate SpeechGraph metrics. Hence, the current study addresses this gap by applying SpeechGraphs analysis to speech produced by individuals with SSD within a joint-task, the STST (Fossard et al., 2018), whose controlled manipulation of referential complexity across stories allows for the examination of its impact on SpeechGraph metrics.

1.2 Objectives and Hypotheses

The overall aim of this study was to use a computational approach to further our understanding of the discourse production impairments exhibited by people with SSD during a task intended to simulate an ecologically valid joint-task conversation.

The computational approach used was SpeechGraphs, in which verbal productions are modeled as graphs capturing the sequential organization and connectivity of the verbal production. We applied it to data from the STST (Fossard et al., 2018). The material of this joint-task consists of nine cartoon stories, each composed of six images that an interaction partner must reorder based on the participant's verbal narration. The STST introduces a manipulation of the referential complexity of the stories, by varying the number of characters in the story and whether or not they share the same gender expression. We have three main objectives:

- (O1) To identify which SpeechGraph metrics are affected in people with SSD relative to non-clinical controls and whether these group effects are influenced by the manipulation of referential complexity.
- (O2) To examine the association between SpeechGraphs metrics and the subjective metric of Facility from Cayouette et al. (2025).
- (O3) To examine whether SpeechGraphs metrics explain unique variance in perceived Facility, beyond what symptoms account for.

2 Method

2.1 Participants

The data used from this study comes from the Banque de données sur la cognition, les interactions et le fonctionnement social (Databank on Cognition, interactions and Social Functioning), a databank hosted at the CIUSSS-CN (ethics number 2013-52/312-2012). Participants from the databank were included in the current study if they completed the STST (Fossard et al., 2018) and were either: (1) a participant with a confirmed diagnosis of a schizophrenia spectrum disorder (SSD; N = 50), or (2) a community control participant (CO; N = 68) that did not present with a mental health condition and was not taking any psychoactive medication at the time of the assessments. Participants were excluded if they presented with a neurological disorder, an intellectual deficit or a physical limitation that could hinder the completion of the tests (e.g. an uncorrected visual impairment) or if they were not fluent in French. More information about both groups of participants is presented in [Supplementary S1](#).

2.2 Procedure Of Data Collection

The STST (Fossard et al. 2018) is a joint task based on the referential communication paradigm (Clark and Wilkes-Gibbs, 1986). The task is performed between the participant and a research assistant who acts as the interaction partner. The material consists of a series of nine cartoon stories, each comprised of six images (without text). At the beginning of each trial, the participant receives one cartoon story, with the images in a predetermined order, while the interaction partner receives the same 6 images composing the story, but in a random order. The order of the nine stories was randomized across participants. During the task, the participant and the interaction partner are separated by an opaque panel so that they cannot see each other (see Figure 2), and the participant is asked to tell the story so that the interaction partner can place the images of the story in the correct order. The interaction partner can provide feedback during the narration of the story, for instance saying "ok", or ask for clarifications if needed. The cartoon stories are divided into three levels of referential complexity, with three level 1 stories with a single character (easy to identify in the discourse), three level 2 stories with two characters of different gender expressions (still relatively easy to distinguish between one another), and three level 3 stories with two characters with the same gender expression (harder to distinguish in the discourse when expressions such as "the girl" or "she" could potentially refer to any of the two female presenting characters) (see Figure 1). The participants are not told about this manipulation of the level of referential complexity. The task is audio recorded.

2.3 Transcription And Data Cleaning

Audio recordings were transcribed verbatim in the context of another project (Achim et al. in preparation). For this study, partner-produced content, including confirmations and clarification requests, was removed from the transcripts, as were disfluencies (e.g. repetitions). Filled pauses (e.g. "um", "uh") were uniformly written as "heu".

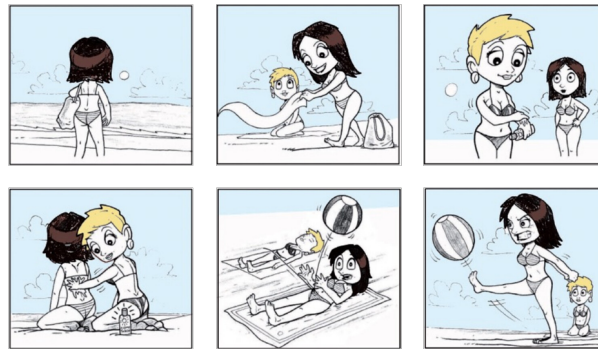
A. example of level 1 : single character**B. example of level 2 : two characters, different gender****C. example of level 3 : two characters, same gender**

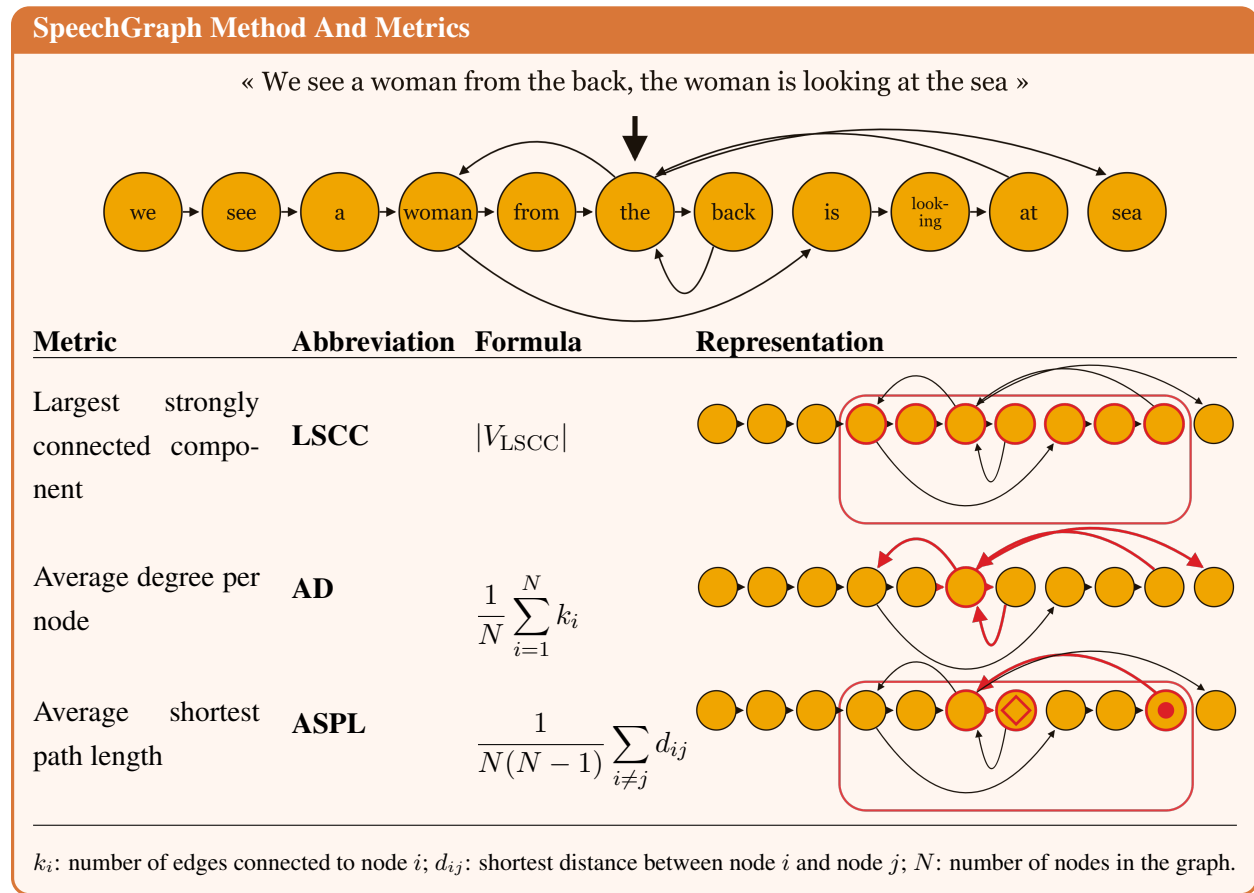
Figure 1: Examples of stories of each referential complexity.



Figure 2: Task setup.

2.4 SpeechGraph Metrics

Following data cleaning, SpeechGraphs were computed for each transcript (i.e. each story) produced by each participant using the NetworkX Python library (Hagberg et al., 2008). Each graph represents spoken language as a directed network in which words are nodes and consecutive word-transitions are edges (Mota et al., 2012). From these graphs, different metrics can be derived, with three main metrics targeted for this study (see Box *SpeechGraph Method And Metrics*). The Largest Strongly Connected Component (LSCC) is the biggest subgraph in which every node can be reached from every other node by following the directed edges. The Average Degree (AD) is the mean number of edges connected to each node across the graph. The Average Shortest Path Length (ASPL) is the mean of the shortest directed distances between all pairs of nodes within the LSCC. These metrics were selected based on their theoretical relevance to the construct of interest (narrative disturbances) and previous research use (Tang et al., 2023; Spencer et al., 2021; Nikzad et al., 2022).



2.5 Statistical Analysis Plan (O1) Group Differences In SpeechGraph Metrics Across Complexity Conditions.

All models were estimated using the Python *statsmodels* library (Seabold and Perktold, 2010). For each metric, a linear mixed-effects model (LMM) was built through the following sequential procedure. All likelihood-ratio tests (LRT) in Steps 1–4 use models estimated by maximum likelihood (ML); Step 5

switches to restricted maximum likelihood (REML) for final inference. Metrics were z-scored across the full sample before any model was fitted.

Step 1: Random-effects structure (LRT_{RE}). Two models sharing the same fixed-effects structure (group + level + group $\times$ level + n_{tokens}) were compared by ML. A model with a random intercept per participant and a random intercept per story, versus the same model augmented with a random slope for the linear level contrast per participant. The slope was retained when the LRT was significant. The random-effects structure determined here was held fixed in all subsequent steps.

Step 2: N -tokens covariate (LRT_{cov}). With the random-effects structure fixed, the necessity of including narrative length as a covariate was assessed by comparing the full model (group + level + group $\times$ level + n_{tokens}) to a reduced model excluding n_{tokens} ($\chi^2(1)$, ML). A significant LRT_{cov} would indicate that production length accounts for non-negligible variance in the metric, justifying its inclusion as a covariate in all subsequent steps. When retained, n_{tokens} was also z-scored across the full sample.

Step 3: Group main effect (LRT_{group}). With the random-effects structure and n_{tokens} covariate fixed (Steps 1–2), a model including the group predictor and the group $\times$ level interaction was compared by ML to a reduced model containing only Level and n_{tokens} ($\chi^2(1)$). By the marginality principle, the group main effect is retained in the model whenever the group $\times$ level interaction is significant (Step 4), irrespective of the LRT_{group} outcome.

Step 4: Group $\times$ Level interaction (LRT_{FE}). The omnibus significance of the interaction term was assessed by comparing the full model (group + level + group $\times$ level + n_{tokens}) to a reduced model without the interaction ($\chi^2(1)$), both estimated by ML with identical random-effects structures. The interaction was retained in the final model when significant.

Step 5: Final model estimation (REML). The model with the fixed- and random-effects structures determined in Steps 1–4 was re-estimated by restricted maximum likelihood (REML) for parameter inference. Fixed effects were: (i) Group (CO = 0, SSD = 1), (ii) a linear orthogonal contrast for difficulty Level (Level 1 = -1, Level 2 = 0, Level 3 = +1), (iii) the Group $\times$ Level interaction (when retained), and (iv) total number of words per transcript as a covariate. Wald z -statistics from the REML model are reported.

Interaction decomposition. For metrics showing a significant Group $\times$ Level interaction, two complementary simple-effect contrasts were derived analytically from the REML fit, the SSD–CO difference at each Level and the linear Level slope within each Group. All contrasts are linear combinations of the existing fixed-effect estimates, with standard errors obtained by the delta method on the fixed-effects covariance matrix.

Multiple comparisons and effect sizes. Given the reduced number of tested pairs and the rationale for each metric, no correction for multiple comparisons was applied to primary inference. Benjamini-Hochberg

false discovery rate (FDR) adjusted results are nonetheless reported. Effect sizes were quantified using Cohen's $f^2 = (R_{\text{full}}^2 - R_{\text{reduced}}^2) / (1 - R_{\text{full}}^2)$, where R^2 denotes the McFadden pseudo- R^2 derived from ML log-likelihoods. For the Group main effect, the reduced model excluded both the group predictor and the interaction term. For the Group $\times$ Level interaction, the reduced model excluded only the interaction term.

Random-effects specification sensitivity. To probe the robustness of the primary inferences to the choice of random-effects structure, the final model was refit under four alternative specifications, ranging from the full structure adopted here to a minimal random-intercept-only model, following the recommendations of [Barr et al. \(2013\)](#).

2.6 Statistical Analysis Plan (O2) Associations between SpeechGraph metrics and Facility.

Behavioral Measure

Facility (*Moyenne facilité totale*) ratings were drawn from the database containing [Cayouette et al. \(2025\)](#) information. For each participant, a naïve listener had rated on a 0–10 scale how easy it was to reorder the images of the story based on the participant's narration, this is what we call Facility, results were aggregated across all stories.

Aggregation and Statistical Analysis

For each participant, the three SpeechGraphs metrics were averaged across all stories. Normality of each variable was assessed with a Shapiro–Wilk test. Pearson's r was used when both variables satisfied the normality assumption, and Spearman's ρ otherwise. Given the reduced number of three tested pairs and the rationale for each metric, no correction for multiple comparisons was applied.

2.7 Statistical Analysis Plan (O3) Unique contribution of SpeechGraph metrics to Facility beyond symptoms.

Symptom Measures

Symptom severity scores were retrieved from the same databank described in the Participants section, where they had been entered as part of the broader clinical assessment of SSD participants. To allow direct comparison with prior work, the symptom dimensions from the Positive and Negative Syndrome Scale (PANSS; [Kay et al. \(1987\)](#)) used in this study are those reported by [Cayouette et al. \(2025\)](#): positive symptoms, negative symptoms, cognitive disorganization, depression/anxiety, excitement/hostility, and the PANSS P2 item (conceptual disorganization).

Commonality Analysis

The relative explanatory contribution of symptoms and SpeechGraph structure was quantified using a commonality analysis (Ray-Mukherjee et al., 2014). This approach decomposes the total explained variance of a regression model into components that are attributable uniquely to each predictor block and to their overlap. Concretely, three nested ordinary least squares (OLS) models were fitted: a symptom-only model (M_{sx} : Facility $\sim$ all symptom dimensions listed above), a SpeechGraph-only model (M_{sg}), and a full model combining both blocks (M_{full}). From the coefficient of determination, R^2 , (i.e., the proportion of variance in the outcome explained by the model) of these three models, four non-overlapping variance components were derived: variance unique to symptoms ($A = R_{full}^2 - R_{sg}^2$, i.e., the proportion of variance that symptoms explain beyond anything SpeechGraph can account for), variance unique to SpeechGraph ($B = R_{full}^2 - R_{sx}^2$, i.e., the proportion that SpeechGraph explains beyond symptoms), shared variance ($AB = R_{sx}^2 + R_{sg}^2 - R_{full}^2$, reflecting the proportion that either block can explain but that cannot be attributed to one alone), and unexplained variance ($U = 1 - R_{full}^2$). The primary comparison of interest was B versus A , as a larger unique SpeechGraph component would indicate that graph structure captures aspects of perceived Facility that symptom severity does not.

The statistical significance and magnitude of the SpeechGraph block's unique contribution were assessed via an F -test comparing M_{full} to M_{sx} , Cohen's f^2 (the ratio of unique SpeechGraphs variance to unexplained variance, B/U), and the squared partial correlation (partial $R^2 = B/(B + U)$), reflecting the proportion of the variance in perceived ease not already accounted for by symptoms that is uniquely explained by the SpeechGraph block.

3 Results

3.1 Preliminary Analysis

Production Length By Group

Total length of production did not differ significantly between groups. A Mann–Whitney U test confirmed no reliable difference in median token counts (SSD $M = 97.2$ vs. CO $M = 104.7$; $U = 1582$, $p = .414$, $d = -0.21$). Linear mixed-effects modeling likewise revealed no significant group effect ($\beta = -7.37$, 95% CI $[-20.29, 5.55]$, $z = -1.12$, $p = .26$) (Figure 3).

Metrics Correlation

LSCC and AD are strongly associated with production length whereas ASPL does not show significant correlation with production length (Figure 4).

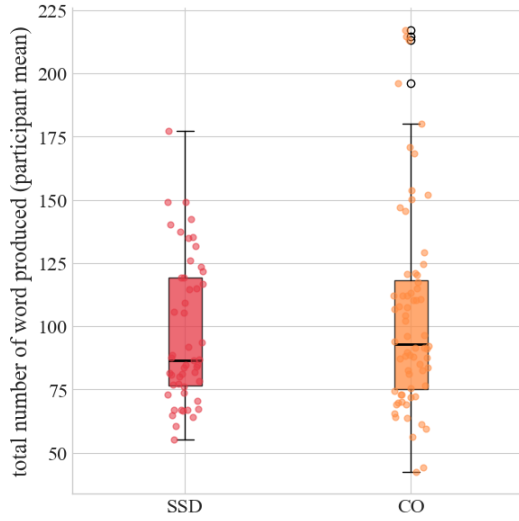


Figure 3: **Distribution of mean number of words counts per participant**, grouped by diagnostic status (SSD vs. CO).

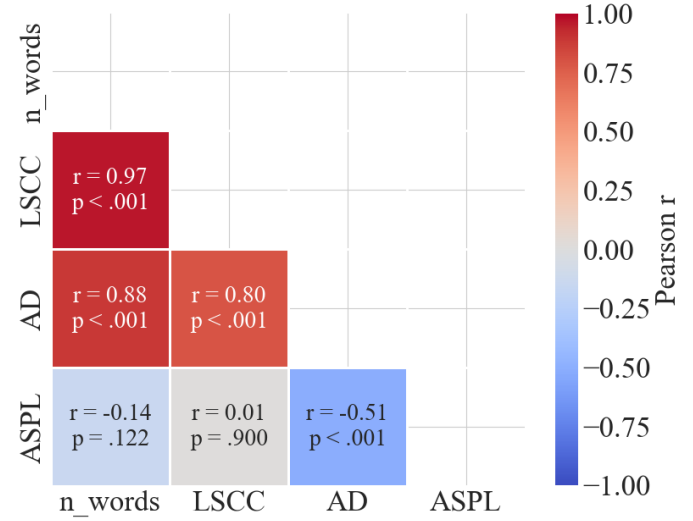


Figure 4: **Lower-triangular correlation heatmap of SpeechGraphs metrics and production length variables (n_words).**

3.2 (O1) Group Differences In SpeechGraphs Metrics Across Complexity Conditions.

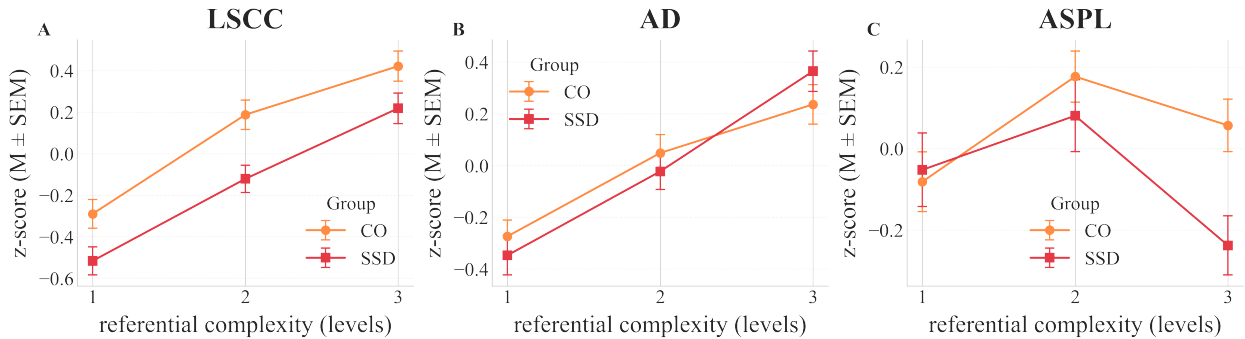


Figure 5: **Evolution of the speech-graph metrics** across the three levels of referential complexity (Level 1, 2, 3), for SSD and CO. Each panel displays group trajectories with error bands.

Figure 5 illustrates how each SpeechGraphs metric evolves across the three levels of referential complexity, separately for SSD and CO groups. Supplementary S2 (Table S2) details the sequential model-building procedure. Step 1 supported the retention of random slopes for Level per participant across all three metrics. Step 2 established that narrative length (n_{tokens}) contributed substantially to model fit for all three metrics, justifying its inclusion as a covariate throughout. Figure 6 illustrates how each SpeechGraphs metric evolves across the three levels of referential complexity, separately for SSD and CO groups, after adjusting for the narrative length (n_{tokens}).

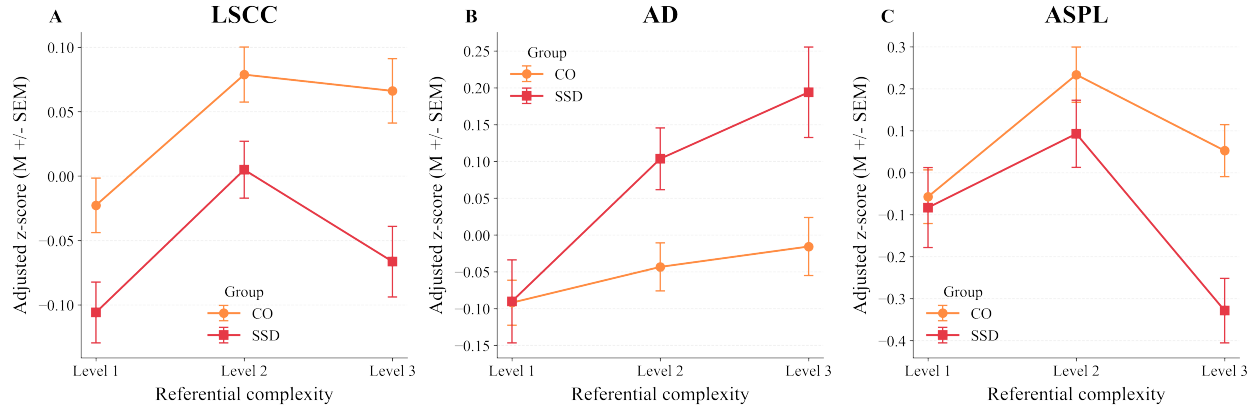


Figure 6: **Group $\times$ Level pattern on the three primary metrics, after adjusting for narrative length.** Each observation has been adjusted by subtracting the linear contribution of n_{tokens} ($y_{\text{adj}} = y - \hat{\beta}_{n_{\text{tokens}}} \cdot n_{\text{tokens}_z}$).

Supplementary S2 (Table S2, Steps 3–4) also presents the model-comparison tests with and without the groups and the interaction between group and referential complexity. A significant Group main effect (SSD vs. CO) was observed on all three primary metrics, with small effect sizes. The Group $\times$ Level interaction was statistically supported for AD and ASPL but not for LSCC (Step 4).

Table 1 reports the REML parameter inferences (Step 5) for Group, Level, and their interaction across the three primary SpeechGraphs metrics. A linear effect of Level on LSCC was also detected, whereas the Level main effects on AD and ASPL did not reach significance after partialling out narrative length, consistent with the strong length-mediation.

Supplementary S2 (Step 6) describes a post hoc LRT examining the joint inclusion of age and sex as demographic covariates, which was non-significant for all three metrics and shifted the group estimates by less than $|0.04|$ standard errors.

Supplementary S3 (Tables S3 and S4) presents the random-effects specification sensitivity analysis, confirming that the Group $\times$ Level interactions for AD and ASPL were stable across all four random-effects structures.

Table 1. Step 5 : final REML estimates for Group, Level, and Group $\times$ Level interaction on speech-graph metrics (controlling for narrative length)

	LSCC	AD	ASPL
Group (SSD vs. CO)			
β	−0.096	+0.119	−0.180
<i>SE</i>	0.037	0.056	0.086
<i>z</i>	−2.585	+2.135	−2.083
<i>p</i>	.010**	.033*	.037*
<i>p</i> _{FDR}	.036*	.047*	.047*
Cohen's f^2	.020	.005	.003
Level (linear)			
β	+0.044	+0.038	+0.055
<i>SE</i>	0.015	0.028	0.050
<i>z</i>	+2.933	+1.354	+1.103
<i>p</i>	.003**	.176	.270
<i>p</i> _{FDR}	.017*	.293	.338
Cohen's f^2	.612	.067	.018
Group $\times$ Level interaction			
β	−0.021	+0.100	−0.168
<i>SE</i>	0.023	0.042	0.075
<i>z</i>	−0.906	+2.363	−2.245
<i>p</i>	.365	.018*	.025*
<i>p</i> _{FDR}	.456	.042*	.042*
Cohen's f^2	.003	.003	.002

* $p < .05$, ** $p < .01$, *** $p < .001$. All DVs z-scored across the full sample; REML estimation. Random-effects structure (random intercept and slope for Level per participant; random intercept per story) selected through the sequential procedure detailed in Supplementary S2. Level coded $\{-1, 0, +1\}$; Group treatment-coded $\{0, 1\}$. The Level slope reported here is therefore the slope for the reference group (CO); the SSD Level slope is $\beta_{\text{Level}} + \beta_{\text{Group} \times \text{Level}}$. Sequential model-building tests (random-effects choice, LRT for fixed effects, covariate sensitivity) reported in Supplementary S2 (Table S2).

Decomposition of the Group $\times$ Level interactions for AD and ASPL.

Table 2 presents the REML estimates for the decomposed interaction effects. To clarify what drives the Group $\times$ Level interactions observed for AD and ASPL, we decomposed each interaction into two complementary sets of simple-effect contrasts derived analytically from the REML fits reported in Table 1. The first set estimates the SSD–CO difference at each Level; the second estimates the linear Level slope within each Group.

For both AD and ASPL, the SSD–CO difference was indistinguishable from zero at Level 1, reached marginal significance at Level 2, and was strongest at Level 3. The complementary slope analysis revealed that the Level effect on these metrics was almost entirely carried by individuals with SSD. For AD, the CO Level slope was non-significant whereas the SSD slope was strongly positive. For ASPL, CO showed no Level slope and SSD showed a non-significant marginal negative slope (Figure 6).

Table 2. Decomposition of the Group $\times$ Level interaction for AD and ASPL

	AD	ASPL
Simple effect of Group at each Level (SSD – CO, z-units)		
<i>Level 1</i>		
β [95% CI]	+0.019 [−0.096, +0.135]	−0.012 [−0.236, +0.212]
p / p_{FDR}	.742 / .742	.914 / .914
<i>Level 2</i>		
β [95% CI]	+0.119 [+0.010, +0.228]	−0.180 [−0.350, −0.011]
p / p_{FDR}	.033* / .049*	.037* / .056
<i>Level 3</i>		
β [95% CI]	+0.219 [+0.063, +0.375]	−0.348 [−0.572, −0.124]
p / p_{FDR}	.006** / .018*	.002** / .007**
Level slope for each Group (z-units per Level step)		
<i>Controls (CO)</i>		
β [95% CI]	+0.038 [−0.017, +0.093]	+0.055 [−0.043, +0.153]
p	.176	.270
<i>SSD</i>		
β [95% CI]	+0.138 [+0.073, +0.202]	−0.113 [−0.227, +0.001]
p	< .001***	.052

* $p < .05$, ** $p < .01$, *** $p < .001$.

3.3 (O2) Associations between SpeechGraph metrics and Facility.

Shapiro–Wilk tests on the participant-level distributions indicated significant non-normality for perceived Facility ($W = .928$, $p < .001$) and LSCC ($W = .946$, $p < .001$), whereas ASPL ($W = .979$, $p = .073$) and AD ($W = .980$, $p = .094$) did not deviate from normality significantly. Because at least one variable in every pair was non-normal (Facility), Spearman rank correlations were retained for all reported associations.

Table 3. Spearman (ρ) correlations between speech-graph metrics and perceived ease of communication by group

	LSCC	AD	ASPL
SSD group ($n = 44$)			
facility ratings	$\rho = .14$	$\rho = .26$	$\rho = -.39^{**}$
CO group ($n = 68$)			
facility ratings	$\rho = .50^{***}$	$\rho = .45^{***}$	$\rho = -.12$

* $p < .05$, ** $p < .01$, *** $p < .001$. Correlations were computed at the participant level using mean metric scores averaged across stories.

Schizophrenia spectrum disorder group

Table 3 displays the Spearman correlations between the three SpeechGraph metrics and perceived Facility for each group. In the SSD group, ASPL was negatively correlated with perceived Facility. While LSCC and AD were not significantly correlated to Facility.

Community controls

In the CO group, both AD and LSCC were strongly and positively correlated with Facility ratings, while ASPL showed no significant association.

3.4 (O3) Unique contribution of SpeechGraph metrics to Facility beyond symptoms

The three metrics constituted the SpeechGraphs block entered into the commonality analysis along with the total number of words per transcript. The variance decomposition (Figure 7) and the residual relationship between speech-graph structure and perceived Facility after controlling for symptoms (Figure 8) are displayed below.

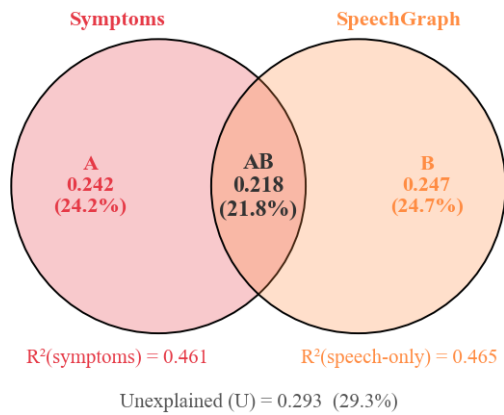


Figure 7: **Variance Decomposition Of Perceived Facility.** Venn diagram decomposing the explained variance in perceived Facility of communication into three components: variance uniquely attributable to symptom severity (A), variance uniquely attributable SpeechGraphs metrics (B), and shared variance (AB)

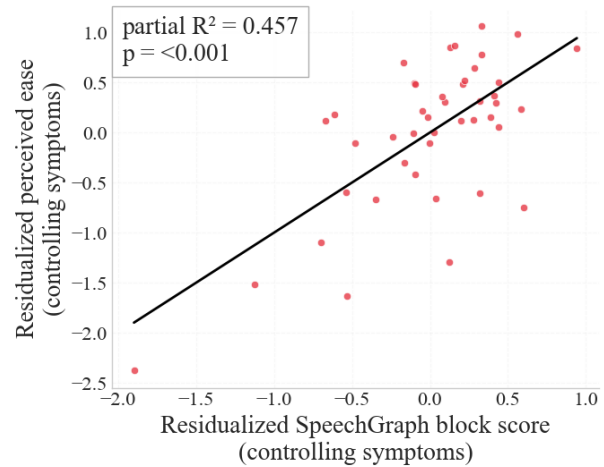


Figure 8: **Partial Regression Of SpeechGraph Unique Variance.** Scatter plot isolating the unique contribution of SpeechGraphs structure to perceived Facility after controlling for symptom severity. The x -axis is the SpeechGraph block score residualised for symptoms; the y -axis is perceived Facility residualised for symptoms.

The commonality analysis revealed that symptom severity and SpeechGraph metrics each explained a comparable and statistically unique share of variance in perceived Facility (Figure 7). Unique contributions were 24.2% for symptoms ($R^2 = .461$ for symptoms alone) and 24.7% for SpeechGraph metrics ($R^2 = .465$ for SpeechGraphs alone), with 21.8% shared between the two blocks. Together they accounted for 70% of total variance, leaving 29.3% unexplained. The unique SpeechGraphs contribution was statistically significant ($F(5, 34) = 5.728, p < .001$; Cohen's $f^2 = 0.842$; partial $R^2 = .457, p < .001$; Figure 8).

4 Discussion

The present study aimed to examine how the structural organisation of discourse production, captured through SpeechGraphs metrics, varies as a function of referential complexity in a joint-task, and how it relates to perceived Facility to achieve a shared goal for naive listeners, in individuals with SSD compared to community controls. Three main findings emerged. First, significant interactions between Group and Referential Complexity were observed on AD and ASPL. As referential demand increased, individuals with SSD produced denser graphs with contracted path structures relative to controls, while controls showed no comparable structural adjustment. Second, the association between SpeechGraphs metrics and perceived Facility differed markedly between groups. In the CO group, perceived Facility was primarily driven by the breadth and connectivity of the SpeechGraph, whereas in SSD it was the contraction of the path structure that most strongly predicted lower perceived Facility. Third, the structural metrics of SpeechGraphs captured a substantial portion of the variance in perceived Facility that was not redundant with aggregated

clinical symptom ratings, supporting the proposal that SpeechGraphs metrics index a distinct dimension of communicative functioning. The remainder of this discussion unpacks each of these findings in turn before considering some limitations of the present study.

4.1 Interpretation of Results

(O1) Group Differences In SpeechGraphs Metrics Across Complexity Conditions

SpeechGraphs of individuals with SSD show significantly smaller LSCC relative to community controls, a result consistent with the literature (Mota et al., 2012, 2017; Palaniyappan et al., 2019; Spencer et al., 2021). Additionally, while cross-linguistic validity of SpeechGraphs had previously been assessed via automatic translation of Portuguese reports into four languages, including French, with preserved classification accuracy (Mota et al., 2014), those analyses did not involve speech produced by native French speakers.

For AD, a significant main effect of Group was observed, with SSD participants showing higher AD than controls when controlling for production length. This finding, however, is most meaningfully interpreted in light of the significant interaction between Group and Referential Complexity. For SSD the AD shows a steeper increase with referential complexity relative to CO. This pattern is consistent with what Mota et al. (2023) describe as short-range recurrence dominance, a structural signature associated with more disrupted narrative organization. It also aligns with classical metrics of lexical diversity. The type-token ratio (amount of distinct words divided by the total amount of words), for instance, has been shown to be specifically reduced in thought-disordered schizophrenia individuals, reflecting a more repetitive and lexically constrained discourse production (Manschreck et al., 1984) that corresponds to the increase of AD.

A significant interaction between Group and Referential Complexity emerged from the ASPL, and this interaction remained stable in the sensitivity analysis excluding the length covariate, suggesting that it is not explained by differences in verbal output. This pattern suggests that, as Referential Complexity increases, the path structure of SSD participants' SpeechGraphs contracts, which means that words become more closely interconnected relative to that of controls. Combined with the AD interaction, this is consistent with a lower lexical diversity adaptation under increasing Referential Complexity. Visually, this pattern is visible in the comparison of Figure 5 and Figure 6, the ASPL is the only metric for which the pattern of interaction is preserved after controlling for production length, an observation coherent with the weak association between ASPL and production length observed in the preliminary analysis.

A closer decomposition of these interactions reinforces the load-revealed reading of the AD and ASPL effects. For both metrics, the SSD–CO difference is null at the simplest narrative condition, and becomes largest and most robust at the highest Referential Complexity. Divergences between groups emerge only as the task imposes greater referential demand. Complementarily, the Level effect itself is carried almost entirely by the SSD group. Controls show no reliable change in AD or ASPL across Levels, whereas SSD participants show a significant increase in AD and a contraction in ASPL.

Taken together, the AD increase and ASPL contraction observed in SSD under high referential complexity

may not be two independent findings but rather a single structural phenomenon projected onto two complementary metrics. Higher mean degree (more edges per node) and shorter average path length both signal a more clumped, compact graph. Under increasing referential demand, SSD discourse appears to collapse onto a less bigger set of semantic targets compared to CO speech, that expands its lexical reach. This convergence reinforces the lexical-narrowing hypothesis formulated above and prepares the ground for the dissociation observed at the level of perceived Facility (O2).

(O2) Associations Between SpeechGraph Metrics And Facility

The relationship between SpeechGraphs metrics and perceived Facility was assessed using Spearman correlation. The correlations differed markedly between groups, to an extent that was unexpected. In the CO group, both the AD and the LSCC were strongly and positively associated with facility ratings, while the ASPL showed no significant association. This suggests that, for CO, larger connected subgraphs and more connected graphs are easier to follow for a naïve listener. In the SSD group, however, the ASPL emerged as the strongest and only predictor of Facility, such that smaller ASPL corresponded to lower perceived Facility. Notably, the primacy of the ASPL in the SSD group converges with the (O1) finding that SSD participants' path structure contracted more under increasing referential complexity.

(O3) Unique Contribution Of SpeechGraphs Metrics To Facility Beyond Symptoms

To examine whether SpeechGraphs structure captured aspects of perceived Facility not already accounted for by clinical symptom severity, a commonality analysis was conducted within the SSD group. The symptom-only model accounted for 46% of variance in perceived Facility, the SpeechGraphs-only model for 47%, and the full model combining both blocks for 71%. Decomposing this explained variance revealed that the proportion uniquely attributable to the SpeechGraph block was comparable to that uniquely attributable to symptom severity. These results indicate that the structural organization of discourse production captures aspects of perceived communicative quality that aggregated symptoms ratings fail to reflect. This is consistent with the proposal that NLP-derived (e.g., SpeechGraphs here) structural metrics may function as biosocial markers of psychosis ([Palaniyappan, 2021](#)). This perspective aligns with phenomenological accounts of social dysfunction in schizophrenia, which argue that the core disturbance is not reducible to observable behavioral deficits or symptom severity. It consists of a qualitative alteration of the capacity for what [Stanghellini and Ballerini \(2002\)](#) named dis-sociality, defined as a disorder of primordial intersubjectivity and common sense. From these standpoints, capturing how the speech of individuals with SSD is received and interpreted by others constitutes a clinically meaningful dimension in its own right.

4.2 Limitations

Sample Composition

A first limitation to report is the same reported by [Cayouette et al. \(2025\)](#). There is an overrepresentation of men compared to women in the SSD group and the SSD group is significantly older than the CO group. Although these demographic imbalances could in principle confound group differences in SpeechGraphs

metrics, a formal likelihood-ratio test comparing models with and without age and sex as covariates yielded non-significant results for all metrics (Supplementary S2, Step 6), and group effect estimates shifted by no more than 0.011 standard deviations upon their inclusion. These results indicate that age and sex do not meaningfully contribute to model fit and do not confound the observed group differences, providing post-hoc justification for their exclusion from the primary models.

Although the PANSS P2 item (Conceptual Disorganization) was included in the commonality analysis, it was not used as a proxy for formal thought disorder severity to examine whether the observed group differences in SpeechGraphs metrics were modulated by individual differences in thought disorder. It could thus be interesting in the future to determine if the greater difficulties performing the tasks with people with SSD are modulated by the presence of formal thought disorder symptoms, and to measure their SpeechGraphs metrics to determine if the same patterns of association with Facility are observed. Especially since past research using Speechgraphs had also specifically focused on formal thought disorders (Mota et al., 2017; Spencer et al., 2021).

Statistical Power

A further limitation concerns statistical power. With $n = 50$ SSD participants and $n = 68$ controls, the present study may be underpowered to detect small-to-medium effects reliably, particularly for interaction terms. The significant interactions between Group and Referential Complexity for AD and ASPL, while surviving FDR correction, should be interpreted with caution, and under the light of the small effect sizes observed across the studies. Replication in larger cohorts is needed before strong inferences can be drawn.

Lack of Comparison in the Literature

While SpeechGraphs metrics have been linked to social cognitive abilities (e.g. emotional processing and mentalization) in SSD (Tang et al., 2023) and functional outcomes (Palaniyappan et al., 2019), the tasks employed preclude direct examination of communicative uptake (e.g. Facility here) as it unfolds during the execution of a shared goal. The present study is the first to examine how SpeechGraphs metrics relate to referential complexity and to perceived Facility.

Sensitivity to Verbatim Preprocessing

A notable sensitivity analysis concerns the preprocessing decision to remove repeated words coming from disfluencies (e.g. "she use- she uses a brush" → "she uses a brush") from the transcripts. Even though Mota et al. (2023) states that there is no need for standardization since the method adopted is "the most naïve point of view to define edges: the relationship between words did not consider any linguistic relationship or rules and was based directly on the sequence of nodes. This decision was taken to avoid subjectivity in the analysis and prevent adjusting the narrative to any normative data or linguistic rule.", studies have handled disfluencies, such as filled pauses, and repetitions inconsistently. In other words, the issue with SpeechGraphs processing is not once the text is transcribed, but on the choices of what should be written,

normalized (e.g. filled pauses in our case), or ignored. Future work should establish shared conventions, particularly as SpeechGraphs method is extended to new languages and ecological paradigms.

Ambiguity in ASPL Computation

Additionally, a related ambiguity concerns the computation of ASPL itself. SpeechGraphs are directed by construction, as edges reflect the temporal order of word transitions, and directed graphs are not guaranteed to be strongly connected (i.e., it is not necessarily possible to reach every node from every other node following the direction of the edges). As illustrated in Box [SpeechGraph Method And Metrics](#), in a graph built from a narrative opening such as “*we see a ...*”, the nodes *we*, *see*, and *a* cannot be reached from any node appearing after *a*. When a graph is not strongly connected, ASPL is undefined for unreachable node pairs, and different implementations handle this case differently: ignoring undefined pairs, substituting infinity, or restricting computation to a connected subgraph. The implementation used by [Nikzad et al. \(2022\)](#) can be identified from their shared code: ASPL was computed using full undirected graph without restricting to a strongly connected subgraph. By default, the coding library employed substitutes $N + 1$ (N being the amount of nodes in the graph) for unreachable node pairs rather than excluding them, which artificially inflates the resulting ASPL values. Whether [Tang et al. \(2023\)](#) adopted the same approach remains unclear, as no implementation details are reported. In the present study, we adopted the principled decision to compute ASPL within the Largest Strongly Connected Component (LSCC), ensuring that the metric reflects path-length properties over a well-defined, fully traversable subgraph, and is not inflated by unreachable pairs. However, direct comparison of our ASPL values with those of [Nikzad et al. \(2022\)](#) is therefore not possible, and comparability with [Tang et al. \(2023\)](#) cannot be established.

4.3 Future Directions

Decomposing Referential Complexity

The present study operationalised referential complexity as a composite manipulation of linear level increase in the LMM model. In the same model, all levels were included, meaning that the transition from one to two characters and the transition from two characters with distinct gender to two characters the same gender expression, were examined in a single model. Future work should disentangle these two dimensions. The effect of number (how many distinct referents must be tracked) and the effect of gender (whether characters share or differ in gender expression, constraining pronoun disambiguation) may have separable consequences structural metrics derived from the SpeechGraphs. Future studies should, however, be aware of the floor effect observed in the present study, as no distinction in speechgraphs metrics was observed for lower levels of referential complexity.

Backtracking Account

In the STST, the interaction partner is allowed to ask for clarifications (e.g. “Could you be more specific?”). These clarification requests lead the participant to go back to a previous part of the story and reformulate it,

this constitutes a backtracking. In conversation analysis, such clarification requests are referred to as other-initiated repair, a mechanism by which a listener signals a problem in hearing, or understanding, and prompts the original speaker to resolve it (Schegloff et al., 1977). Those repairs in the task, associated with the results from Cayouette et al. (2025) (the lower Facility scores results) open the way for future analysis. For instance, investigating how backtracking behaviors differs between SSD and community participants, and how it relates to SpeechGraphs metrics and perceived Facility. Do SSD participants engage in less backtracking? Are backtracking episodes from individuals with SSD more disorganized, leading to more disconnected SpeechGraphs and lower facility ratings? Understanding the role of backtracking could provide deeper insights into the mechanisms underlying communicative disturbances and repairs in SSD.

Global versus local SpeechGraphs metrics

Several previous studies used the SpeechGraphs method in window of words (e.g. 30 words) and computed local metrics on these windows, which were then averaged across the entire narrative (Nikzad et al., 2022; Palaniyappan et al., 2019; Mota et al., 2014). By contrast, the present study focused on global graph metrics that capture overall structural properties of the SpeechGraphs. We choose this approach since the STST task manipulates referential complexity at the level of the entire narrative: For the stories with two characters, the one in focus on the first two images, will come back as the focus of images 5 & 6, while the second character will be the focus of the images 3 & 4. This means that the referential complexity manipulation is expected to impact the overall structure of the graph, and thus global metrics were considered more appropriate to capture this effect. That being said, exploratory analyses using local metrics could still provide complementary insights into the current work, especially around backtracking events.

4.4 Conclusion

Overall, this study provides novel insights into the structural organization of discourse production in individuals with SSD, and its relationship to communicative uptake (e.g. Facility) in a joint task. The observed interactions between group and referential complexity on AD and ASPL suggest that individuals with SSD exhibit a distinct pattern of structural adjustment under increasing referential complexity demand. The dissociation between groups in the predictors of perceived Facility further indicates that the structural features of discourse production that facilitate communication differ between individuals with SSD and non-clinical populations. Finally, the unique contribution of SpeechGraphs metrics to perceived Facility beyond clinical symptoms underscores the potential of computational structural assessments to capture clinically meaningful dimensions of communicative functioning in SSD.

Taken together, the findings of this study suggest that computational structural assessments of discourse production in joint tasks can provide a richer characterization of the communicative disturbances in SSD. One that is sensitive to the mechanisms of structural organization underlying discourse production impairments. Which, moreover, carries direct implications both for identifying the communicative barriers that hinder social recovery, and for pointing to what to look for as early language disturbances across the onset and course of the disorder.

Declarations

Ethics

The study was approved by the Comité d'éthique de la recherche sectoriel en neurosciences et santé mentale du CIUSSS-CN (project 2424-2831).

Data Availability

The data supporting the findings of this study cannot yet be made publicly available due to ethical considerations. Qualified researchers may contact the supervisor of this work for access.

Notebooks and code for the data analysis are available on the following github repository:

https://github.com/EvannR/SpeechGraph_Rabeau_master_thesis

Declaration of AI use

- Copilot on visual studio code was used to assist the writing of the code for the data analysis.
- QWEN3.6 27B was used to a limited extent for the preprocessing of the data, it was run locally as the clinical data must not be shared with third parties. Its use was limited to the following preprocessing steps: 1) removing disfluencies and repetitions from the transcripts, 2) correcting typos in the transcripts, 3) standardizing the format of the transcripts for input into the SpeechGraph pipeline.

Following the principles outlined by [Knöchel et al. \(2025\)](#), no identifiable data was submitted to the GenAI tool used for code development, and human oversight was maintained over all analyses.

Conflict of Interest

The author received honoraria for a conference presentation on NLP applications in psychiatric data - including the present method - at the University of Neuchatel. The author has no other financial or non-financial conflicts of interest to declare.

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Declaration Of Contribution

	Evann Rabeau	Amélie Achim	DISCOURSE group	Research team
Status	Master's student	Principal Investigator	DISCOURSE	Researchers, post-docs, PhD students
Topic	Decision	Decision	Discussion	Discussion
Research question	Proposal and decision	Proposal and decision	Inputs	
Literature review to define the methods	Did the review and wrote the synthesis	Referenced key papers and proofread the review	Referenced key papers	Suggested references
Implementation of the experiment	Implemented the SpeechGraph pipeline and transcript preprocessing	Validated the implementation		
Analysis plan	Did the analysis plan	Recommendations on analyses	Recommendations on analyses	
Data analysis	Made the code (SpeechGraph metrics, LMM, commonality analysis)	Methodological feedback		
Interpretation of the results	main interpretation	proofreading	Discussion	
Writing of the thesis	Writing	Review		

Pre-registraion

The pre-registration document asked by the Master program at the beginning of the internship period was written by the author about another project and another lab, no pre-registered document exist for the work presented in this thesis.

Supplementary Materials

S1. Participant Characteristics

Table S1. Socio-demographic and clinical characteristics of participants

	SSD group (<i>n</i> = 50)	CO group (<i>n</i> = 68)	statistics
socio-demographic information			
age (mean, s.d.)	35.2 (11.6)	28.5 (9.0)	$t(116) = 3.54, p < .001$
sex			
men (N)	37	34	$\chi^2(1) = 5.96, p = .015$
women (N)	13	34	
duration of illness (years; mean, s.d.)	9.8 (10.8)		
diagnosis (N)			
schizophrenia	27		
schizoaffective disorder	12		
delusional disorder	4		
psychotic disorder NOS	4		
brief psychotic disorder	1		
schizophreniform disorder	2		
PANSS scores per dimension (mean, s.d.)			
positive dimension (/42)	12.4 (5.5)		
negative dimension (/49)	15.0 (6.6)		
cognitive/disorganization (/35)	10.1 (3.2)		
depression/anxiety (/28)	8.0 (2.8)		
excitability/hostility (/28)	5.5 (2.1)		
PANSS total score (/182)	51.0 (14.5)		
P2. Conceptual disorganization (/7)	1.8 (1.0)		
principal antipsychotic (N)			
clozapine	13		
quetiapine XR	11		
aripiprazole	12		
risperidone	3		
olanzapine	2		
paliperidone	2		
lurasidone	1		
brexpiprazole	1		
none	5		
COST (mean, s.d.)			
ToM items (/52)	37.9 (7.8)	45.5 (3.6)	$t(116) = 7.08, p < .001$

Note. PANSS = Positive and Negative Syndrome Scale; P2 = item P2 of the PANSS (conceptual disorganization); COST = Competitive Social Task; ToM = Theory of Mind. Statistics: independent samples *t*-tests (two-tailed); χ^2 = Pearson chi-squared test.

S2. Sequential Model-Building Procedure

The primary linear mixed-effects models for O1 were specified through a sequential model-building procedure. Steps 1–4 and 6 use likelihood-ratio tests (LRT) under maximum-likelihood (ML) estimation for direct comparison of nested models.

Table S2. Sequential model-building likelihood-ratio tests for O1

	LSCC	AD	ASPL
Step 1: Random slope retention (LRT_{RE}, ML)			
$\chi^2(3)$	202.430	90.436	45.534
<i>p</i>	< .001***	< .001***	< .001***
Decision	Retained	Retained	Retained
Step 2: <i>N</i>-tokens covariate necessity (LRT_{cov}, ML)			
$\chi^2(1)$	1908.44	894.81	37.37
<i>p</i>	< .001***	< .001***	< .001***
β_{group} (no cov.)	−0.212	+0.011	−0.150
Step 3: Group main-effect block (LRT_{group}, ML)			
$\chi^2(2)$	6.847	7.669	9.094
<i>p</i>	.033*	.022*	.011*
<i>p</i> _{FDR}	.041*	.036*	.026*
Step 4: Group × Level interaction (LRT_{FE}, ML)			
$\chi^2(1)$	0.834	5.546	5.025
<i>p</i>	.361	.019*	.025*
<i>p</i> _{FDR}	.451	.042*	.042*
Decision	Not retained	Retained	Retained
Step 6: Age + Sex covariate necessity (LRT_{demo}, ML)			
$\chi^2(2)$	3.071	1.497	1.605
<i>p</i>	.215	.473	.448
$\Delta\beta_{\text{group}}$	−0.026	−0.004	−0.000

Note. **p* < .05, ***p* < .01, ****p* < .001. All LRTs use ML estimation.

S3. Random-Effects Specification Sensitivity

Each primary metric was refit under four random-effects specifications, holding the fixed-effects formula (Group, Level, $\text{Group} \times \text{Level}$, n_{tokens} covariate) and REML estimation constant. The specifications were: Full (random intercept and slope for Level per participant, random intercept per story; the primary model); No story RI (without the story random intercept); No participant RS (without the per-participant random slope for Level); and Minimal (random intercept per participant only). Results for Group and Level main effects are reported in Table S3; $\text{Group} \times \text{Level}$ interaction results are reported in Table S4.

Removing the story random intercept (No story RI) left all fixed-effect estimates, standard errors, and p -values unchanged to three decimal places across all three metrics.

The $\text{Group} \times \text{Level}$ interaction estimate for AD ranged from $\beta = +0.100$ to $\beta = +0.105$ across the four specifications (p between .010 and .018). For ASPL, the estimate ranged from $\beta = -0.168$ to $\beta = -0.177$ (p between .015 and .025). Both interactions were statistically significant in all four specifications. The interaction was non-significant for LSCC across all specifications.

Dropping the by-participant random slope (No participant RS) reduced the standard error on the Group estimate by 40–50% across all three metrics (LSCC: $0.037 \rightarrow 0.019$; AD: $0.056 \rightarrow 0.035$; ASPL: $0.086 \rightarrow 0.061$), with corresponding p -values falling to $\leq .003$ (Barr et al., 2013).

Table S3. Random-effects specification sensitivity: Group and Level main effects (REML)

Group (SSD vs. CO)	Full	No story RI	No part. RS	Minimal
<i>LSCC</i>				
β	-0.096	-0.096	-0.091	-0.097
<i>SE</i>	0.037	0.037	0.019	0.037
<i>p</i>	.010**	.010**	< .001***	.009**
<i>AD</i>				
β	+0.119	+0.119	+0.115	+0.118
<i>SE</i>	0.056	0.056	0.035	0.055
<i>p</i>	.033*	.033*	.001**	.033*
<i>ASPL</i>				
β	-0.180	-0.180	-0.178	-0.180
<i>SE</i>	0.086	0.087	0.061	0.086
<i>p</i>	.037*	.037*	.003**	.037*
Level (linear)				
<i>LSCC</i>				
β	+0.044	+0.044	+0.034	+0.047
<i>SE</i>	0.015	0.015	0.016	0.013
<i>p</i>	.003**	.004**	.034*	< .001***
<i>AD</i>				
β	+0.038	+0.038	+0.050	+0.042
<i>SE</i>	0.028	0.028	0.029	0.026
<i>p</i>	.176	.175	.082	.113
<i>ASPL</i>				
β	+0.055	+0.055	+0.049	+0.055
<i>SE</i>	0.050	0.050	0.049	0.046
<i>p</i>	.270	.270	.317	.238

Note. * $p < .05$, ** $p < .01$, *** $p < .001$. All models share the same fixed-effects formula (Group, Level, Group $\times$ Level, n_{tokens} , z -scored) and REML estimation. **Full**: random intercept per participant, random slope for Level per participant, random intercept per story (primary model). **No story RI**: drops the by-story random intercept. **No participant RS**: drops the per-participant random slope for Level. **Minimal**: random intercept per participant only. The columns “Full” and “No story RI” are identical to three decimal places across all rows, reflecting that the by-story variance component is estimated at ≈ 0 in this sample. The Group $\times$ Level interaction is reported in Table S4.

Table S4. Random-effects specification sensitivity: Group $\times$ Level interaction (REML)

Group $\times$ Level interaction	Full	No story RI	No part. RS	Minimal
<i>LSCC</i>				
β	−0.021	−0.021	−0.025	−0.021
<i>SE</i>	0.023	0.023	0.024	0.020
<i>p</i>	.365	.365	.294	.290
<i>AD</i>				
β	+0.100	+0.100	+0.105	+0.101
<i>SE</i>	0.042	0.042	0.043	0.039
<i>p</i>	.018*	.018*	.015*	.010**
<i>ASPL</i>				
β	−0.168	−0.168	−0.177	−0.169
<i>SE</i>	0.075	0.075	0.074	0.069
<i>p</i>	.025*	.025*	.017*	.015*

Note. * $p < .05$, ** $p < .01$, *** $p < .001$. Specifications and conventions as in Table S3. The interaction estimate and its significance remain stable across all four random-effects structures, including the Minimal specification (random intercept per participant only).

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